

Inspiring Young Minds
Through Knowledge Olympiads

CLASS

6

SET-B

DO NOT OPEN THIS BOOKLET UNTIL ASKED TO DO SO

Roll No.:

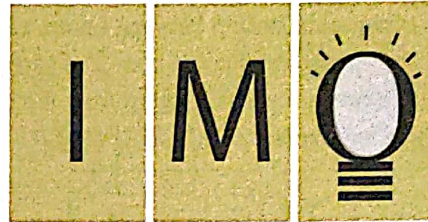
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Total Questions: 50

Time: 1 hr.



**9TH SOF INTERNATIONAL
MATHEMATICS OLYMPIAD**

Guidelines for the Candidate

1. You will get additional ten minutes to fill up information about yourself on the OMR Sheet, before the start of the exam.
2. The Question Paper comprises four sections: **Logical Reasoning** (15 Questions), **Mathematical Reasoning** (20 Questions), **Everyday Mathematics** (10 Questions) and **Achievers Section** (5 Questions)

Each question in Achievers Section carries 3 marks, whereas all other questions carry one mark each.

3. All questions are compulsory. There is no negative marking. Use of calculator is not permitted.
 4. Write your **Name**, **Roll No.** and **Class** clearly on the **OMR Sheet** and do not forget to sign it.
 5. There is only ONE correct answer. Choose only ONE option for an answer.
 6. To mark your choice of answers by darkening the circles in the OMR Sheet, use an **HB Pencil** or a **Blue / Black ball point pen** only. E.g.
- Q. 16: A number has nine ones, six tens and eight hundreds. What is the number?
- | | |
|--------|--------|
| A. 869 | B. 896 |
| C. 968 | D. 986 |

As the correct answer is option A, you must darken the circle corresponding to option A in the OMR Sheet.

16. ● (B) (C) (D)

7. Rough work should be done in the blank space provided in the booklet.
8. Return the OMR Sheet to the invigilator at the end of the exam.
9. Write your Roll No. on the Question Paper too and take it home for future reference.

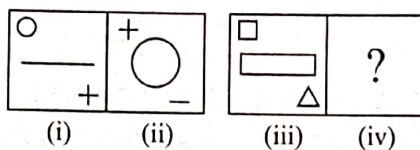
LOGICAL REASONING

1. How many such symbols are there in the given arrangement, each of which is immediately preceded by a number and immediately followed by a consonant?

R 5 M E % 4 W 1 A 2 D # K \$ 3 P 9 @ F B © 8 J
I 7 * H 6 Q V Y

- A. One
- B. Two
- C. Three
- D. More than three

2. There is a certain relationship between figures (i) and (ii). Establish a similar relationship between figures (iii) and (iv) by selecting a suitable figure from the given options which will replace the (?) in figure (iv).



- A.
- B.
- C.
- D.

3. In a certain code language, ENGLISH is written as FMHKJRI. How is OCTOBER written in that code?

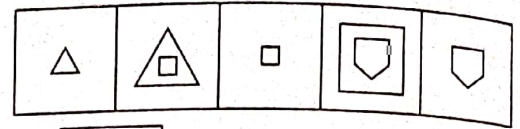
- A. PBUNCDS
- B. PBUCNSD
- C. BPUNCSD
- D. PBUCNDS

4. Pravin walked 30 metres towards East, took a right turn and walked 20 metres, again took a right turn and walked 30 metres. How far is he from his starting point?

- A. 30 metres
- B. 80 metres
- C. 50 metres
- D. 20 metres

5. Select a figure from the options which will continue the same series as established by the Problem Figures.

Problem Figures



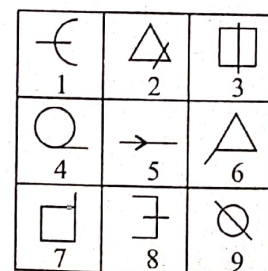
- A.
- B.
- C.
- D.

6. Which number will be the smallest, if the first and the last digits of each of the following numbers are interchanged?

389 476 635 847 568

- A. 389
- B. 476
- C. 635
- D. 847

7. Arrange the given figures into three classes using each figure only once.



- A. 1, 3, 9 ; 2, 5, 8 ; 4, 6, 7
- B. 1, 5, 8 ; 4, 6, 7 ; 2, 3, 9
- C. 2, 5, 9 ; 1, 3, 8 ; 2, 6, 7
- D. 1, 8, 9 ; 4, 6, 7 ; 2, 3, 5

8. Which of the following options will complete the pattern in Fig. (X)?

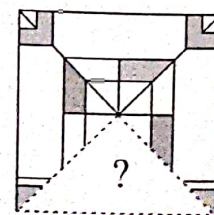
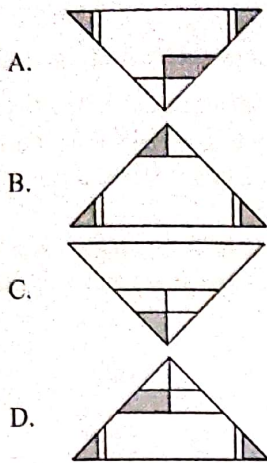


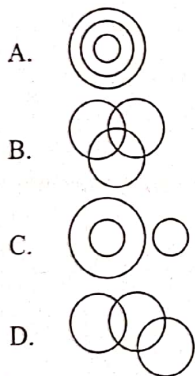
Fig. (X)



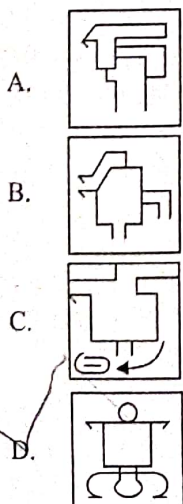
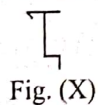
9. Introducing a man, a woman says, "His wife is the only daughter of my father." How is the man related to the woman?

- A. Brother
- B. Father-in-law
- C. Brother-in-law
- D. Husband

10. Which of the following Venn diagrams best represents the relationship amongst "Fruit, Mango, Grass"?



11. In which of the following figures, Fig. (X) is exactly embedded as one of its part?

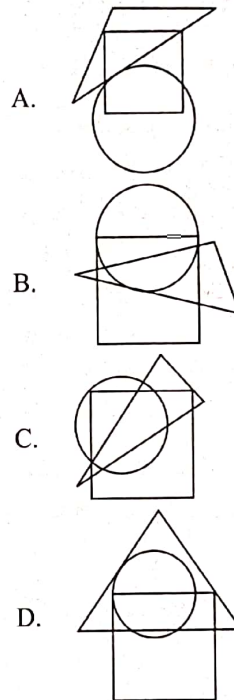


12. Find the missing number, if a certain rule is followed row-wise or column-wise.

- A. 6
- B. 7
- C. 8
- D. 9

3	6	8
5	5	4
6	9	?

13. Which of the following options satisfies the same condition of placement of the dots as in Fig. (X)?



14. How many meaningful English words can be formed from the letters A, I, P, R using each letter only once?

- A. Six
- B. Five
- C. Four
- D. None of these

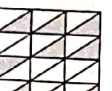
15. Select the correct mirror image of the given combination, if mirror is placed vertically to the right.

● M 2 T h 5 F 8 I ●

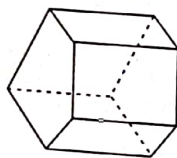
- A. ● W 5 L P 2 E 8 I ●
- B. ● I 8 F 2 d T 5 M ●
- C. ● W 5 L P 5 E 8 I ●
- D. ● M 8 F 2 d T 5 I ●

MATHEMATICAL REASONING

16. Select the INCORRECT match of equivalent shaded fraction given in Column-I and Column-II.

Column-I	Column-II
A. 	
B. 	
C. 	
D. 	

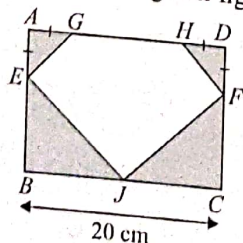
17. How many faces does the given solid have?



- A. 5
B. 6
C. 7
D. 10
18. If 8 pens cost ₹ w , find the cost of 5 such pens.

- A. ₹ $40w$
B. ₹ $\left(\frac{5w}{8}\right)$
C. ₹ $\left(\frac{80w}{5}\right)$
D. ₹ $\left(\frac{w}{40}\right)$

19. The perimeter of rectangle $ABCD$ is 76 cm. AB is 3 times as long as AE . BC is twice as long as BJ . Find the shaded area in the given figure.



- A. 132 cm^2
B. 148 cm^2
C. 150 cm^2
D. 156 cm^2

20. Which of the following statements is CORRECT?
- (i) A sector is the region in the interior of a circle enclosed by an arc on one side and a pair of radii on the other two sides.
- (ii) A segment of a circle is the region in the interior of the circle enclosed by an arc and a chord.

- A. Both (i) & (ii)
B. Only (i)
C. Only (ii)
D. Neither (i) nor (ii)

21. Which of the following statements is INCORRECT with respect to the whole numbers?

- A. They are closed under addition and multiplication.
B. Division by 0 is not defined.
C. Addition and subtraction are commutative.
D. Multiplication is distributive over addition.

22. If the number $517*324$ is exactly divisible by 3, then the smallest whole number in place of $*$ will be

- A. 0
B. 1
C. 2
D. 3

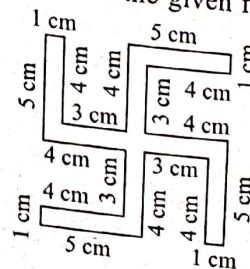
23. If $47.2506 = 4A + \frac{7}{B} + 2C + \frac{5}{D} + 6E$, then the value of $5A + 3B + 6C + D + 3E$ is

- A. 53.6003
B. 53.603
C. 153.6003
D. 213.0003

24. Which sign will come in the box to make the expression true?

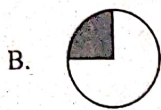
- $(-25) - (-42) - (-27) \quad \square \quad (-42) - (-25) + (-22)$
- A. $<$
B. $>$
C. $=$
D. $\leq$

25. Find the perimeter of the given figure.



- A. 17 cm
B. 68 cm
C. 43 cm
D. 96 cm

26. In which of the following figure does the shaded part represents 0.3 ?



27. 80325901 can be written in words as

- A. Eight crore thirty two lakh five thousand nine hundred one
B. Eight crore three lakh twenty five thousand nine hundred one
C. Eighty lakh thirty two thousand nine hundred one
D. Eighty crore thirty two lakh five thousand nine hundred one

28. In case of a scalene triangle _____.

- A. Two angles are equal
B. All the angles are equal
C. All the angles are of different measures
D. None of these

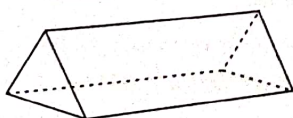
29. Simplify : $4\frac{3}{5} - 2\frac{7}{9} - 1\frac{2}{15} - \frac{2}{5}$

- A. $\frac{14}{45}$
B. $\frac{13}{45}$
C. $\frac{16}{45}$
D. $\frac{12}{45}$

30. HCF of an even number and an odd number is always

- A. 1
B. 0
C. 2
D. Can't say

31. The given figure is called as _____ which contains _____ faces ; _____ corners and _____ edges.



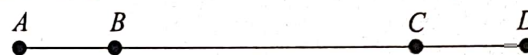
- A. Triangular prism, 5, 6, 9
B. Triangular prism, 6, 9, 5
C. Rectangular prism, 5, 9, 6
D. Rectangular prism, 9, 6, 5

32. Read the given statements below and identify the correct option.

- (i) A parallelogram has only one line of symmetry.
(ii) A square and a rectangle have the same line of symmetry.
(iii) The diameter of a circle is its line of symmetry.

- | | (i) | (ii) | (iii) |
|----|-------|-------|-------|
| A. | False | True | False |
| B. | True | False | True |
| C. | False | False | True |
| D. | False | False | False |

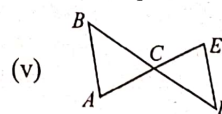
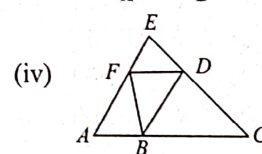
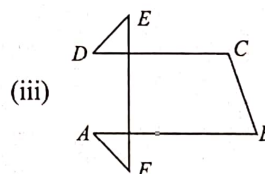
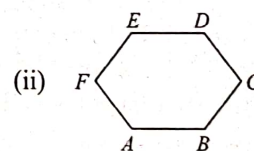
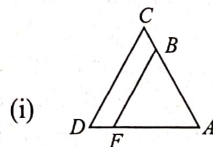
33. On a straight road there are four cities A, B, C and D.



Distance between B and D is 39 km, between A and C is 27 km and between C and D is 15 km. How far is A from B?

- A. 2 km
B. 3 km
C. 4 km
D. 5 km

34. Which of the following figures are NOT polygons?



- A. (i), (ii) and (v)
B. (i), (ii) and (iii)
C. (i), (iii), (iv) and (v)
D. (i), (ii), (iii) and (iv)

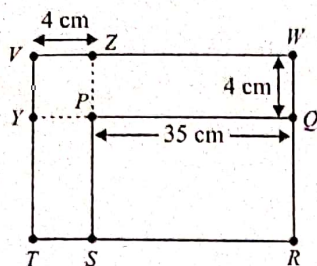
35. The greatest number that exactly divides 105, 1001 and 2436 is

- A. 3
B. 7
C. 11
D. 21

36. A rectangular courtyard 3.78 m long and 5.25 m wide is to be paved exactly with square tiles, all of same size. Then the largest size of the tile which could be used for the purpose is $(n \times 3)$ cm. Find n .
- 6
 - 8
 - 7
 - None of these
37. A courier man covers 35 m distance to deliver a product to the customer. He travels 0.028 km by bicycle and the rest on foot. What distance does he cover on foot?
- 5 m
 - 7 m
 - 8 m
 - 10 m
38. From a barrel containing 513.76 kg of rice, the cook used 27.895 kg in the first three days of the week. How much rice was left in the barrel?
- 386.865 kg
 - 475.685 kg
 - 468.786 kg
 - 485.865 kg
39. An auditorium has a capacity of 2000 seats. There are 200 seats each allotted for ₹ 500 tickets, 500 seats each for ₹ 250 tickets, 600 seats each for ₹ 100 tickets and the rest for ₹ 50 tickets each. If all the tickets are sold, how much money would be collected?
- ₹ 1,60,000
 - ₹ 2,80,050
 - ₹ 3,20,000
 - ₹ 2,20,000
40. Three bells ring at intervals of 30, 45 and 60 minutes respectively. If they begin ringing together at 5 p.m., then they will ring together again at _____.
- 8:30 p.m.
 - 5:30 p.m.
 - 8 p.m.
 - 6 p.m.
41. Arvind has a box of chocolates which is $\frac{7}{8}$ full. Suraj has a box which is $\frac{4}{5}$ full. Whose box has more chocolates and by how much?
- Arvind, $\frac{3}{40}$
 - Suraj, $\frac{3}{40}$
 - Arvind, $\frac{9}{10}$
 - Suraj, $\frac{9}{10}$
42. A boat covers a distance of 15 km in 6 hours. What distance will it cover in 24 hours?
- 32 km
 - 48 km
 - 56 km
 - 60 km
43. Sunil had ₹ 30. He spent ₹ 6.25 on books and ₹ 12.75 on toys. How much amount is left with him?
- ₹ 10
 - ₹ 11
 - ₹ 12.02
 - ₹ 9.02
44. The cost of making a fence around a square field is ₹ 392 at the rate of ₹ 14 per metre. What is the length of each side of the field?
- 7 m
 - 8 m
 - 8 cm
 - 7 cm
45. 32 bags of sugar, each weighing 42 kg, cost ₹ 26880. What is the cost of 25 bags of sugar each weighing 31 kg?
- ₹ 17,535
 - ₹ 19,325
 - ₹ 15,500
 - ₹ 14,850

ACHIEVERS SECTION

46. PQRS is a rectangle. When the length and breadth of the rectangle increases, the rectangle is enlarged to VWRT and the area of the rectangle is increased by 226 cm^2 . If the length of rectangle PQRS is twice its breadth, Find :
- Perimeter of rectangle PQRS.
 - Area of rectangle VWRT.



- (a) (b)
- A. 105 cm 388.5 cm²
 B. 105 cm 838.5 cm²
 C. 108 cm 838.5 cm²
 D. 108 cm 388.5 cm²

47. Which of the following statements is INCORRECT?

- A. Two friends ate $\frac{2}{3}$ and $\frac{1}{4}$ of a chocolate. The fraction of the chocolate they eat together is $\frac{11}{12}$.
 B. A point A on a mountain is 6,450 m above the sea level. Another point B is in a mine 48,600 m below the sea level. The distance between these two points is 55050 cm.
 C. Monika's present age is x years. Her father's age is 5 years more than 3 times her age. Monika's father's age is $(3x + 5)$ years.
 D. On subtracting, the sum of 34.29 and 158.3 from the sum of 21.947 and 201.3, we get 30.657.

48. Study the statements carefully.

Statement I : Any number is divisible by 5, if the sum of the digits of the number is divisible by 5.

Statement II : Any number is divisible by 6, if it is divisible by either 2 or 3 or both 2 and 3.

Which of the following options hold?

- A. Both Statement I and Statement II are true.
 B. Both Statement I and Statement II are false.
 C. Statement I is true but Statement II is false.
 D. Statement I is false and Statement II is true.

49. Match the columns.

Column I

Column II

- (i) Number of line(s) passing through two given points is (p) Circumference
 (ii) Number of ray(s) can be drawn with same initial point is (q) Zero
 (iii) The minimum number of point of intersection of three lines (if lines are parallel) is (r) One
 (iv) The distance around a circle is its (s) Infinite
- A. (i) $\rightarrow$ (r), (ii) $\rightarrow$ (s), (iii) $\rightarrow$ (q), (iv) $\rightarrow$ (p)
 B. (i) $\rightarrow$ (r), (ii) $\rightarrow$ (p), (iii) $\rightarrow$ (q), (iv) $\rightarrow$ (s)
 C. (i) $\rightarrow$ (s), (ii) $\rightarrow$ (q), (iii) $\rightarrow$ (p), (iv) $\rightarrow$ (r)
 D. (i) $\rightarrow$ (s), (ii) $\rightarrow$ (r), (iii) $\rightarrow$ (q), (iv) $\rightarrow$ (p)

50. The total number of students who applied for a scholarship in the last five years is given below.

Year	2008	2009	2010	2011	2012
No. of students	3,000	3,500	4,000	4,500	5,000

Using the symbol $\star$ to represent 500 students, answer the following questions.

- (i) How many symbols are needed to represent the number of students in 2010 ?
 (ii) How many more symbols are needed to represent the number of students in 2012 than in 2009?
- (i) (ii)
- A. 7 4
 B. 7 3
 C. 8 4
 D. 8 3

SPACE FOR ROUGH WORK

12
-5
7

"SOF International General Knowledge Olympiad" - A tournament of minds for all round development of students - is proposed to be launched from the academic year 2016-17.

In association with
 BRITISH COUNCIL



**15TH SOF NATIONAL
CYBER OLYMPIAD**



**18TH SOF NATIONAL
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